Bootstrapping time series data

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The bootstrap: introduction

- The **bootstrap** is a method for approximating the sampling distribution of estimators and test statistics. It is based on **resampling**: repeatedly drawing artificial data sets.
- Bootstrapping is an alternative to asymptotic delta method inference. However, it is also only valid in the limit $T \to \infty$.
- The bootstrap helps us compute critical values and confidence intervals w/o having to plug into complicated asymptotic formulas.
- Sometimes the bootstrap may also deliver **asymptotic refinements**: The bootstrap approx'n to the sampling distribution may be more accurate for small T than the asymptotic approximation.
- There are many versions of the bootstrap, requiring different assumptions. Any bootstrap procedure can fail.
- References: Horowitz (2001), Kilian & Lütkepohl (2017, ch. 12).

Approximating sampling distributions

- We observe data $y_1, \ldots, y_T$ drawn from some true distribution F_0.
- From the data we compute a **statistic** $\hat{Q}_T = \hat{Q}_T(y_1, \ldots, y_T)$. E.g., $\hat{Q}_T = \sqrt{T}(\hat{\theta} - \theta_0)$, where $\hat{\theta}$ is an estimator of θ_0.
- In order to test hypotheses or construct confidence intervals, we would like to know the CDF of the (repeated experiments) sampling distribution of $\hat{Q}_T$:

$$G_T(\tau, F_0) \equiv P_{F_0}(\hat{Q}_T \leq \tau), \quad \tau \in \mathbb{R}.$$

The subscript "F_0" indicates that the probability is wrt. the true distribution of the data.

Approximating sampling distributions (cont.)

- So far in this course we have relied on large-sample approximations: If $\hat{Q}_T \xrightarrow{d} Q$, then
$$G_T(\tau, F_0) \to G_\infty(\tau, F_0),$$
where $G_\infty(\cdot, F_0)$ is the CDF of the limiting random variable Q, the distribution of which depends on F_0.
- There are two concerns with large-sample approximations:
  1. It can be hard to compute quantiles/moments of the limiting distribution $G_\infty(\cdot, F_0)$. E.g., lots of derivatives involved for extremum estimators.
  2. Large-T approximations may work poorly for small T. There is no one-size-fits-all guide to how large T needs to be – depends on skewness, kurtosis, persistence, nonlinearities, etc.

Approximating sampling distributions (cont.)

- Suppose for a moment that we *know* F_0. Then an obvious strategy for approximating the *exact* sampling distribution $G_T(\cdot, F_0)$ of $\hat{Q}_T$ is by **Monte Carlo simulation**:

1. For each $b = 1, \ldots, B$:

   i) Simulate a data set $y_1^*, \ldots, y_T^*$ from F_0.

   ii) Compute the statistic $\hat{Q}_{T,b}^*$ on the simulated data set. E.g., for statistic $\hat{Q}_T = \sqrt{T}(\hat{\theta} - \theta_0)$, compute estimator $\hat{\theta}_b^*$ on the simulated data, and set $\hat{Q}_{T,b}^* \equiv \sqrt{T}(\hat{\theta}_b^* - \theta_0)$.

2. Approximate $G_T(\cdot, F_0)$ by the empirical distribution function (EDF) of $\hat{Q}_{T,b}^*$ across the B simulated data sets:

$$G_T(\tau, F_0) \approx \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{\hat{Q}_{T,b}^* \leq \tau\}, \quad \tau \in \mathbb{R}.$$

- The above approximation becomes exact in the limit as we increase the computational effort $B \to \infty$ (the Glivenko-Cantelli Theorem).

Approximating sampling distributions (cont.)

- In reality we don't know F_0. But we can substitute an estimate $\hat{F}$ of F_0 into the Monte Carlo procedure.
- For example, suppose the true data distribution $F_0(\cdot) = F(\cdot, \theta_0)$ depends on finite-dimensional parameter θ_0. Given a consistent estimator $\hat{\theta}$ of θ_0, we approximate $F_0 \approx \hat{F} \equiv F(\cdot, \hat{\theta})$:

1. For each $b = 1, \ldots, B$:
   i) Simulate a data set $y_1^*, \ldots, y_T^*$ from $\boxed{\hat{F} \equiv F(\cdot, \hat{\theta})}$.
   ii) Compute the statistic $\hat{Q}_{T,b}^*$ on the simulated data set. E.g., for statistic $\hat{Q}_T = \sqrt{T}(\hat{\theta} - \theta_0)$, compute estimator $\hat{\theta}_b^*$ on the simulated data, and set $\boxed{\hat{Q}_{T,b}^* \equiv \sqrt{T}(\hat{\theta}_b^* - \hat{\theta})}$.
2. $G_T(\cdot, F_0) \approx$ EDF of $\hat{Q}_{T,b}^*$ across the B simulated data sets.

Parametric bootstrap

- The algorithm on the previous slide is called the **parametric bootstrap**. It requires:

  1. The data distr'n $F_0(\cdot) = F(\cdot, \theta_0)$ is known up to a param. vector $\theta_0 \in \Theta$.
  2. We can estimate θ_0 consistently with $\hat{\theta}$.
  3. We can simulate $(y_1^*, \ldots, y_T^*) \sim F(\cdot, \hat{\theta})$.
  4. We can compute $\hat{Q}_T^*$ quickly on each simulated data set.

- As $B \to \infty$, the bootstrap EDF of $\hat{Q}_{T,b}^*$ tends to

$$\begin{aligned} P^*(\hat{Q}_T^* \leq \tau) &\equiv P_{F(\cdot,\hat{\theta})}(\hat{Q}_T^* \leq \tau \mid y_1, \ldots, y_T) \\ &= G_T(\tau, F(\cdot, \hat{\theta})), \quad \tau \in \mathbb{R}. \end{aligned}$$

- The notation "P^*" indicates that the probability is taken wrt. the bootstrap distr'n of $\hat{Q}_T^*$, *conditional* on the original data $y_1, \ldots, y_T$.

Parametric bootstrap (cont.)

- The only difference between the parametric bootstrap and the infeasible Monte Carlo computation of $G_T(\cdot, F_0)$ is that the parametric bootstrap substitutes $\hat{F} = F(\cdot, \hat{\theta})$ in place of the unknown data distribution $F_0(\cdot) = F(\cdot, \theta_0)$.
- If $\hat{\theta} \approx \theta_0$, then we expect $\hat{F} \approx F_0$, and thus we expect the bootstrap approximation (for large B)

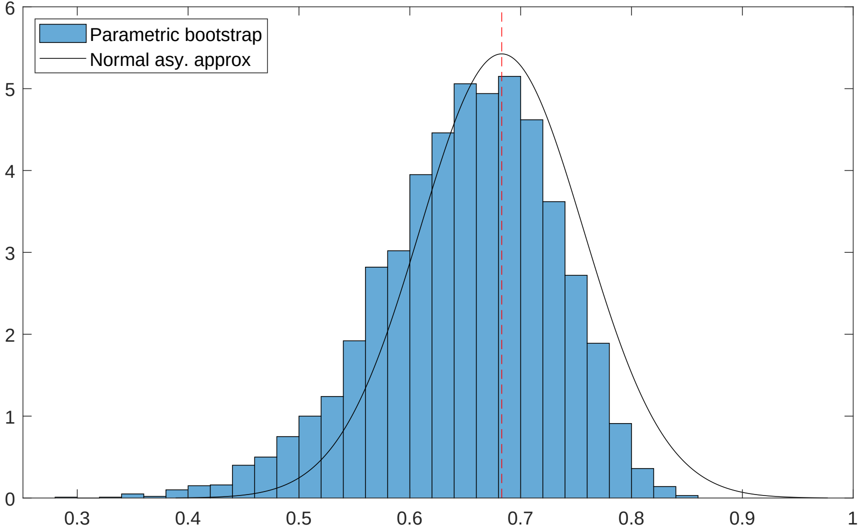
$$G_T(\tau, F(\cdot, \hat{\theta}))$$

to be close to the target sampling distribution $G_T(\tau, F_0)$.

Example: AR(1)

```
T = 100;
rho = 0.7;
sigma = 1;

y = filter(1, [1 -rho], sigma*randn(T,1));
[rhohat,sigmahat] = ar1_estim(y);

B = 5000;
rhohatboot = zeros(B,1);

for b=1:B
    ystar = filter(1, [1 -rhohat], sigmahat*randn(T,1));
    rhohatboot(b) = ar1_estim(ystar);
end
```

Example: AR(1) (cont.)



Nonparametric bootstrap

- Typically, we do not fully know the true data distribution F_0 up to a finite-dimensional parameter θ_0.
- E.g., VAR(p): We know neither the parameters nor the exact distribution of the innovations.
- If the data $(y_1, \dots, y_T)$ were i.i.d., we could still approximate F_0: Use the marginal EDF of the data,

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{y_t \leq y\}, \quad y \in \mathbb{R}.$$

  The **nonparametric bootstrap** approximates F_0 by T independent draws from $\hat{f}(\cdot)$ (EDF of the data).
- Notice that simulating an i.i.d. data set $y_1^*, \dots, y_T^*$ from the EDF $\hat{f}$ is the same as **random sampling with replacement** from the T data points $y_1, \dots, y_T$. Usually the bootstrap data sets will have duplicates, e.g., $y_3^* = y_{17}^* = y_8$.

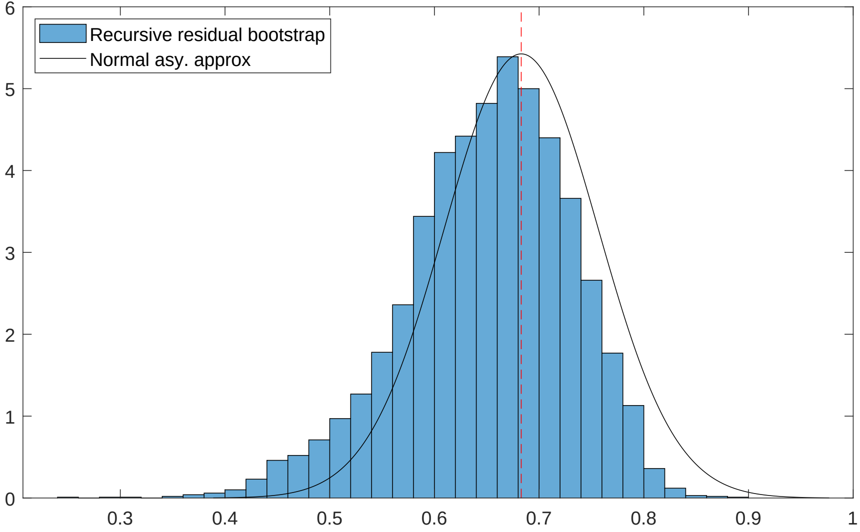
Recursive residual bootstrap

- Unfortunately, time series data is rarely i.i.d., so F_0 will not be well approximated by T independent draws from the marginal EDF $\hat{f}(\cdot)$.
- But many time series models contain *innovations* $\{u_t\}$ which we are willing to assume are i.i.d.
- This suggests applying the nonparametric bootstrap to the model residuals $\{\hat{u}_t\}$, obtained after estimating the model parameters $\hat{\theta}$.
- **Recursive residual bootstrap**:
  1. Sample bootstrap residuals $u_1^*, \ldots, u_T^*$ by random sampling with replacement from the original residuals $\hat{u}_1, \ldots, \hat{u}_T$.
  2. Construct bootstrap data $y_1^*, \ldots, y_T^*$ from the bootstrap residuals $u_1^*, \ldots, u_T^*$ by applying the model's defining equations and substituting $\theta_0 = \hat{\theta}$. For VARs: $y_t^* = \hat{\nu} + \sum_{\ell=1}^{p} \hat{A}_\ell y_{t-\ell}^* + u_t^*$.

Example: AR(1)

```
T = 100;
rho = 0.7;
sigma = 1;

y = filter(1, [1 -rho], sigma*randn(T,1));
[rhohat,sigmahat,uhat] = ar1_estim(y);

B = 5000;
rhohatboot = zeros(B,1);

for b=1:B
    ustar = uhat(ceil(rand(T-1,1)*(T-1)));
    ystar = filter(1, [1 -rhohat], [0; ustar]);
    rhohatboot(b) = ar1_estim(ystar);
end
```

Example: AR(1) (cont.)



Recursive residual bootstrap (cont.)

- For VARs and similar models, we have to take a stand on what to do about the initial observations.
- The choice will not matter for large T under stationarity.
- The previous AR(1) example simply set $y_1^* = 0$, but this is awkward.
- Two choices are often used in practice:

  1. Set $y_t^* = y_t$ for $t = 1, \ldots, p$. That is, condition on the observed pre-sample observations.
  2. Draw $(y_1^*, \ldots, y_p^*)$ as a block of p observations uniformly at random from the $T - p + 1$ possible blocks

$$(y_1, \ldots, y_p), (y_2, \ldots, y_{p+1}), \ldots, (y_{T-p+1}, \ldots, y_T)$$

     in the data. We draw a new initial block at random in every bootstrap repetition.

Bootstrap consistency

- When does the bootstrap "work"?
- A bootstrap strategy is said to be **consistent** if

$$\sup_{\tau} |G_T(\tau, \hat{F}) - G_\infty(\tau, F_0)| \xrightarrow{p} 0 \quad \text{as } T \to \infty,$$

  where $\hat{F}$ is the approximation to the true distribution of the data used in the bootstrap. Here we are assuming $B = \infty$, so there is no *numerical* error in this idealized bootstrap, only *statistical* error.
- Intuitively, we require the bootstrap distribution $\hat{F}$ to mimic those features of the data distribution F_0 that determine the asymptotic distribution $G_\infty(\cdot, F_0)$ of our statistic $\hat{Q}_T$. For example:
  - If the asy. limit of $\hat{Q}_T$ depends on 2nd moments of the data, the bootstrap must mimic those 2nd moments.
  - If serial correlation doesn't matter for the limit distribution, we don't need to mimic the serial correlation of the data.

Reminder: Asymptotic distribution of VAR estimator

- VAR(p) with martingale difference sequence (MDS) innovations:

$$y_t = \beta x_t + u_t, \quad E(u_t \mid \{u_s\}_{s<t}) = 0_{n\times 1},$$

where $x_t \equiv (1, y'_{t-1}, \dots, y'_{t-p})'$ and $\beta \equiv (\nu, A_1, \dots, A_p)$.

- Least-squares estimators of β and $\Sigma \equiv \text{Var}(u_t)$:

$$\hat{\beta} \equiv (\textstyle\sum_{t=1}^T y_t x'_t)(\sum_{t=1}^T x_t x'_t)^{-1}, \quad \hat{\Sigma} \equiv \frac{1}{T}\sum_{t=1}^T (y_t - \hat{\beta}x_t)(y_t - \hat{\beta}x_t)'.$$

- Under some conditions (Brüggemann, Jentsch & Trenkler, 2016),

$$\sqrt{T}\begin{pmatrix} \text{vec}(\hat{\beta}-\beta) \\ \text{vec}(\hat{\Sigma}-\Sigma) \end{pmatrix} \overset{d}{\to} N\left(0, \begin{pmatrix} (S_x^{-1}\otimes I_n)\Omega(S_x^{-1}\otimes I_n) & 0 \\ 0 & \Xi \end{pmatrix}\right),$$

where $S_x \equiv E(x_t x'_t)$, $\Omega \equiv E(u_t u'_t \otimes x_t x'_t)$, and $\Xi \equiv \text{LRV}(u_t \otimes u_t)$ (and thus depends on 4th moments of u_t).

- Note: The limit for $\hat{\beta}$ is the same as in cross-sectional multivariate linear regression where (y_t, x_t) is i.i.d. (even though here it's not).

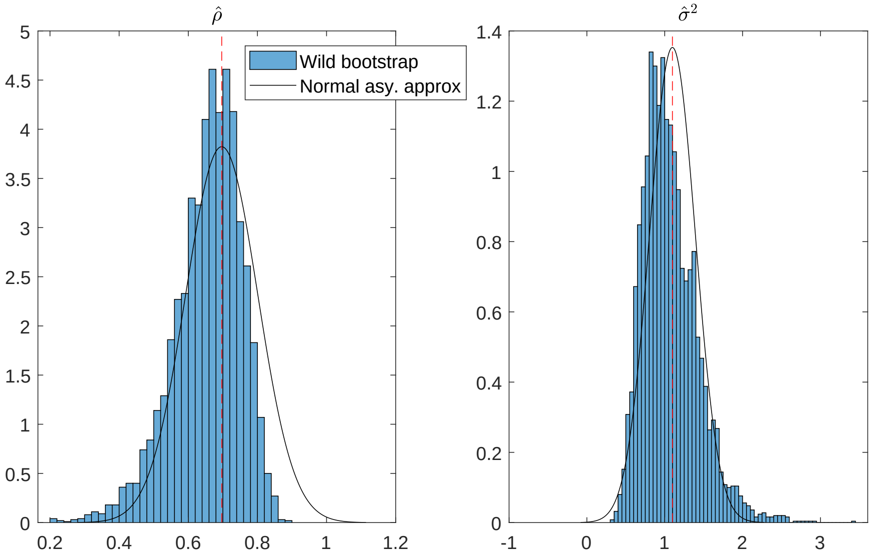
Wild bootstrap

- The recursive residual bootstrap works only with i.i.d. u_t.
- The **wild bootstrap** (WB) allows for heteroskedasticity: $\text{Var}(u_t \mid \{y_s\}_{s<t}) \neq \text{const}$.
- It does require $E(u_t \mid \{u_s\}_{s<t}) = 0_{n\times 1}$, so u_t must be serially uncorr.
- The WB works exactly like the recursive bootstrap, except the residuals u_t^* are generated as $u_t^* \equiv \zeta_t^* \hat{u}_t$ for $t = 1, \dots, T$.
  - ζ_t^* are simulated i.i.d. scalar rv's with mean 0 and variance 1, e.g., $\zeta_t^* \sim N(0,1)$. Note: $\dim(\zeta_t^*) = 1$ even if $n > 1$.
  - Note: There is no sampling with replacement. We simply multiply the original residuals by the variables ζ_t^*, $t = 1, \dots, T$.
- The WB is consistent for $\hat{\beta}$, since it preserves the *cross-second moments* $E[(u_t u_t' \otimes x_t x_t')]$ btw. $x_t = (1, y_{t-1}', \dots, y_{t-p}')'$ and u_t.
- The WB is *inconsistent* for $\hat{\Sigma}$, since limit distribution depends on 4th moments of u_t, which are affected by $\zeta_t^* \sim N(0,1)$.

Example: AR(1) with conditional heteroskedasticity

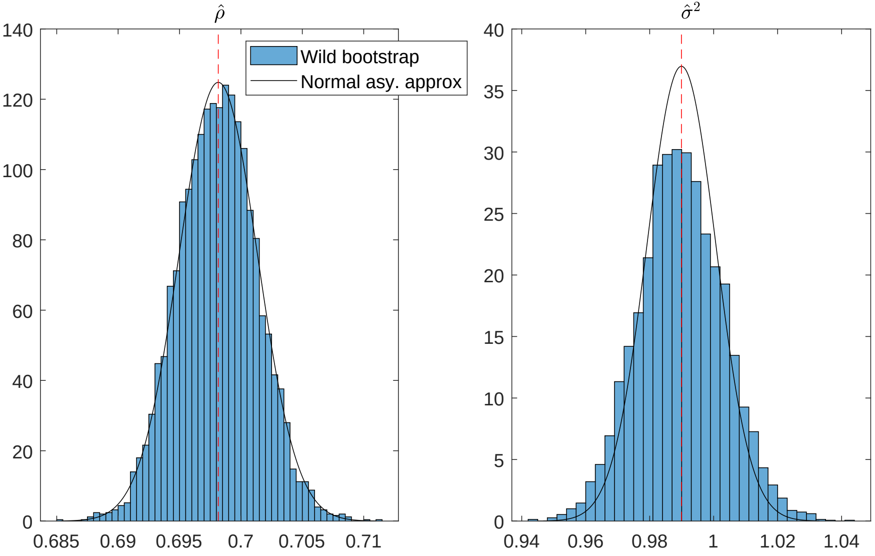
```
T = 100;
rho = 0.7;

epss = randn(T+1,1);
y = filter(1, [1 -rho], abs(epss(1:T)).*epss(2:T+1));
[rhohat,sigmahat,uhat] = ar1_estim(y);

B = 5000;
rhohatboot = zeros(B,1);
sigmahatboot = zeros(B,1);

for b=1:B
    ustar = randn(T-1,1).*uhat;
    ystar = filter(1, [1 -rhohat], [0; ustar]);
    [rhohatboot(b),sigmahatboot(b)] = ar1_estim(ystar);
end
```

Example: AR(1) with conditional heteroskedasticity (cont.)



$T = 100$

Example: AR(1) with conditional heteroskedasticity (cont.)



$T = 100{,}000$

Block bootstrap

- All the bootstrap strategies so far relied on innovations being serially uncorrelated. The **block bootstrap** handles general serial correlation.

- We sample T/m *blocks* of m successive data points, i.e., sample randomly with replacement from the rows of the matrix
$$\begin{pmatrix} y_1' & y_2' & \cdots & y_m' \\ y_2' & y_3' & \cdots & y_{m+1}' \\ & \vdots & & \\ y_{T-m+1}' & y_{T-m+2}' & \cdots & y_T' \end{pmatrix}.$$
Then we stack the T/m blocks in a $T \times n$ matrix to create a bootstrap sample of $\{y_t\}$.

- Bootstrap consistency requires $m, T \to \infty$, $m/T \to 0$.

- By applying the block bootstrap to the innovations $\hat{u}_t$, it can be used to bootstrap $\hat{\Sigma}$ in VARs with conditional heteroskedasticity. See Brüggemann, Jentsch & Trenkler (2016).

Bootstrap confidence interval

- Suppose we bootstrap an estimator $\hat{\theta}$ satisfying

$$\hat{Q}_T \equiv \sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} Q.$$

- Let $\tilde{q}^*_\alpha$ be the α-th sample quantile of the bootstrap draws $\hat{Q}^*_{T,b} \equiv \sqrt{T}(\hat{\theta}^* - \hat{\theta})$, $b = 1, \ldots, B$. Then if the bootstrap is consistent (and $B = \infty$),

$$\lim_{T \to \infty} P(\tilde{q}^*_{\alpha/2} \leq \hat{Q}_T \leq \tilde{q}^*_{1-\alpha/2}) = 1 - \alpha.$$

- Insert $\hat{Q}_T = \sqrt{T}(\hat{\theta} - \theta_0)$ and rearrange the inequalities to obtain:

$$\lim_{T \to \infty} P(\hat{\theta} - T^{-1/2}\tilde{q}^*_{1-\alpha/2} \leq \theta_0 \leq \hat{\theta} - T^{-1/2}\tilde{q}^*_{\alpha/2}) = 1 - \alpha.$$

- Conclusion: the **percentile bootstrap confidence interval**

$$\widehat{\mathrm{CI}} \equiv \left[\hat{\theta} - T^{-1/2}\tilde{q}^*_{1-\alpha/2}, \hat{\theta} - T^{-1/2}\tilde{q}^*_{\alpha/2}\right]$$

is an asy. valid $1 - \alpha$ confidence interval for θ_0.

  - Not a typo: The upper and lower quantiles are "swapped".

Bootstrap confidence interval (cont.)

$$\widehat{\mathrm{CI}} \equiv \left[\hat{\theta} - T^{-1/2}\tilde{q}^*_{1-\alpha/2}, \hat{\theta} - T^{-1/2}\tilde{q}^*_{\alpha/2}\right]$$

- Since $\tilde{q}^*_\alpha$ is the quantile of $\sqrt{T}(\hat{\theta}^* - \hat{\theta})$ across the bootstrap distribution (conditional on the actual data), we have

  $$\tilde{q}^*_\alpha = \sqrt{T}(q^*_\alpha - \hat{\theta}),$$

  where q^*_α is the α-th sample quantile of the bootstrap draws $\hat{\theta}^*_b$, $b = 1, \ldots, B$.
- Hence, we can equivalently write the percentile bootstrap CI as

  $$\widehat{\mathrm{CI}} = \left[2\hat{\theta} - q^*_{1-\alpha/2}, 2\hat{\theta} - q^*_{\alpha/2}\right].$$
- The CI is not necessarily symmetric. Its midpoint is

  $$\hat{\theta} - \underbrace{\left[(q^*_{\alpha/2} + q^*_{1-\alpha/2})/2 - \hat{\theta}\right]}_{\text{implicit }\textbf{bias correction}\text{!}}.$$

Efron bootstrap confidence interval

- If the asy. distribution Q of $\hat{Q}_T$ is symmetric around zero (e.g., normal), and the bootstrap is consistent, then

$$\sqrt{T}(\hat{\theta} - q^*_{\alpha/2}) \approx \sqrt{T}(q^*_{1-\alpha/2} - \hat{\theta})$$

with high probability as $T \to \infty$.

- Inserting this approximation into the percentile CI, we get the **Efron bootstrap confidence interval**

$$\widehat{\text{CI}}_{\text{Efron}} \equiv \left[q^*_{\alpha/2}, q^*_{1-\alpha/2}\right].$$

- Looks intuitively pleasing and is very popular in practice.

- But could have less accurate coverage than the percentile CI if the finite-sample distribution of $\hat{\theta}$ is asymmetric.

Asymptotic refinements

- The bootstrap allows us to avoid complicated delta method formulas.
- Additionally, *sometimes* the bootstrap delivers **asymptotic refinements**: Its approximation error as a function of T tends to zero faster than for the delta method approximation.
- We only get asy. refinements if our statistic $\hat{Q}_T$ is **asymptotically pivotal**, i.e., its asy. distr'n does not depend on unknown param's.
  - Example: Any statistic with asy. $N(0,1)$ or χ^2_r distribution, for known r.
- For intuition, consider the extreme case where $\hat{Q}_T$ is pivotal in *finite samples*, i.e., the finite-T distr'n does not depend on unknown parameters. Then the parametric bootstrap delivers the *exact* finite-T sampling distribution, since for pivotal statistics it doesn't matter that we substitute $\hat{\theta}$ for θ_0.
- See Horowitz (2001) for rigorous details (i.i.d. case).

Percentile-t bootstrap confidence interval

- Suppose $\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \sigma^2)$ and we have a consistent estimator $\hat{\sigma}^2$. Then $\hat{Q}_T \equiv \sqrt{T}(\hat{\theta} - \theta_0)/\hat{\sigma}$ is asy. pivotal.
- Generate bootstrap draws $\tilde{Q}^*_{T,b} \equiv \sqrt{T}(\hat{\theta}^*_b - \hat{\theta})/\hat{\sigma}^*_b$, where $\hat{\theta}^*_b$ and $\hat{\sigma}^*_b$ are computed from each bootstrap data set.
- Let $\tilde{q}^*_{t,\alpha/2}$ and $\tilde{q}^*_{t,1-\alpha/2}$ be the sample quantiles of $\tilde{Q}^*_{T,b}$.
- By similar calculations as for the percentile CI, the following **percentile-t bootstrap CI** is asy. valid:

$$\widehat{\mathrm{CI}}_t \equiv \left[\hat{\theta} - T^{-1/2}\hat{\sigma}\tilde{q}^*_{t,1-\alpha/2}, \hat{\theta} - T^{-1/2}\hat{\sigma}\tilde{q}^*_{t,\alpha/2}\right].$$

- Refinements: The under-coverage rate of the percentile-t CI tends to zero faster (as a fct of T) than for the percentile or Efron CIs.

Simulation evidence

- Kilian (1998) finds that standard bootstrap confidence intervals for impulse responses can have poor small-sample coverage probability. This is ameliorated by using bias-corrected VAR estimates when generating the bootstrap data.
- For a summary of simulation evidence on the performance of different VAR bootstrap strategies, see Kilian & Lütkepohl (2017), ch. 12.5.

When does the bootstrap fail?

- The bootstrap can fail for two reasons:

  1. We fail to mimic the features of the data distribution that are important for the limiting distribution of our statistic.
     - Example: Don't capture 4th moments of u_t correctly when bootstrapping $\hat{\Sigma}$ (VAR residual variance).

  2. The asy. distribution of the statistic depends discontinuously on the data distribution.
     - Example: Bootstrapping the distribution of $\hat{Q}_T \equiv \max_t\{y_t\}$.

- Point 2 is not a concern if our statistic is asymptotically equivalent with a smooth transformation of sample moments (e.g., GMM with smooth moment function).

- See Horowitz (2001) for various examples of bootstrap failure.

- Remember: Bootstrap inference is valid *asymptotically*. For small T, performance could be bad (perhaps worse than delta method).